

Problem Set 6: Statistical Tests, Confidence Intervals and Comparison of Two Samples (April 12 – April 25, 18:00)

Problem 1 (Soil Sample)

We assume that the lead content X in a soil sample is normally distributed, $X \sim \mathcal{N}(\mu, \sigma^2)$.

- a) The standard deviation is known ($\sigma = 10$ ppb (parts per billion)), and we are interested in the expected value μ . The lead content is measured in 10 soil samples, resulting in an average of 31 ppb. Give a 99% confidence interval for μ .

Hint: The cumulative distribution function of the standard normal distribution satisfies $\Phi(2.58) = 0.995$.

- b) How many observations are necessary to reduce the width of the confidence interval by half?
- c) How many independent determinations of the lead content need to be planned if with 99% confidence, it is to be determined to within 1 ppb, that is, if the width of the 99% confidence interval should not be greater than 2 ppb?

Problem 2 (Coin Toss Predictor)

Lisa has built a machine that records the flipping of a coin and predicts the outcome ("heads" or "tails") based on a complex video analysis.

However, she believes that this is purely random, and the machine's predictions are correct only half of the time on average (or worse). Her friend Laura thinks she is a pessimist and wants to prove in a statistical test that the machine doesn't just make random predictions.

For a test run, she flips a coin 20 times and lets the machine predict the outcome. In 13 out of 20 tosses, the machine predicts the correct result. Let X be the number of correct guesses made by the machine (out of $n = 20$ trials). To model X , we use a binomial distribution with parameters $n = 20$ and success probability p , i.e., $X \sim \text{Bin}(n, p)$.

- a) State the null and alternative hypotheses for the statistical test. Justify your choice.
- b) Provide the distribution of X under H_0 . Is the null hypothesis rejected at the 1% significance level?
- c) What is the probability of rejecting the null hypothesis if the true prediction probability is $p = 0.75$?

Problem 3 (Trial)

In a trial, an individual is accused of a crime.

- a) Which wrong judgment corresponds to a Type I statistical error? (Consider the null hypothesis to be the accused's innocence.)

A law student is interested in sentences passed on young adults convicted of serious knife crime in Switzerland. To do that, she randomly selects a sample of 26 cases from judicial files from 2023. She finds that the average sentence length is 6.87 months with a standard deviation of 2.2 months.

- b) Construct a 95%-confidence interval for the length of the prison sentence given to a young adult convicted of a serious knife crime.
- c) What assumptions did you made for part b)?
- d) Suppose that all such convicts in fact served one-half of their sentences. What are the average and standard deviation of the time spent in prison for his random sample of 26 cases?
- e) Test the hypothesis that the average time spent in prison is at least 5 months for such young adults, at level 1%.

Problem 4 (Sample Size)

We consider $X_1, \dots, X_n$, n i.i.d samples from a Bernoulli distribution with parameter $p \in [0, 1]$. We want to test the following Hypothesis:

$$H_0 : p = 0.49 \text{ vs. } H_1 : p = 0.51$$

What should be n so that the probability of a type I error and a type II error are smaller than 0.03?

Hint: Use the Central Limit Theorem.

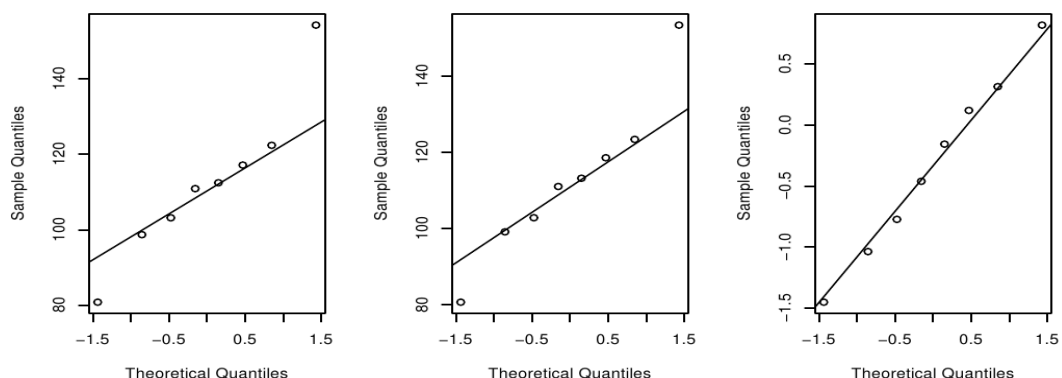
Problem 5 (Downhill Race)

In 2025 the men's downhill race will take place at the Alpine Ski World Championships in Saalbach. Each nation is allowed to start 4 skiers. The coach of the Swiss downhill team has already nominated 3 athletes. For the last remaining starting place, he cannot decide between skier A and skier B. In foresight, he had already announced at the beginning of the season that in such cases, it should be tested whether the seasonal results achieved by the World Championships are statistically significantly different or not. if yes, the skier for which the data speaks shall start; otherwise, a fair coin decides. The times achieved by the two skiers in the previous 8 season downhill races (in minutes:seconds) are as follows:

Race No. (i)	1	2	3	4	5	6	7	8
Skier A (a_i)	1:52.43	2:02.37	1:20.78	1:38.67	2:34.23	1:57.13	1:43.14	1:50.88
Skier B (b_i)	1:53.20	2:03.40	1:20.66	1:39.13	2:33.41	1:58.58	1:42.82	1:51.04

Now a statistical test should be developed to check whether one of the two skiers is significantly faster than the other one. This should be done in such a way that if the two skiers are equally fast on average, the probability that the test decides in favor of one of them is at most 10%.

- a) Is the sample paired or unpaired? Is the test one-sided or two-sided? What is the level α ? With which probability does it come to a coin toss decision if both athletes are equally fast on average?
- b) The following graphs are QQ plots of the samples $(a_1, \dots, a_8)$, $(b_1, \dots, b_8)$, and $(d_1, \dots, d_8)$ (in seconds), where we denote the sample difference by $d_i := a_i - b_i$ $i = 1, \dots, 8$.



Is the assumption of normal distribution justified for the data relevant to the test? Provide a reasoning. Which data set is well described by a normal distribution? Which test is therefore most appropriate?

- c) Develop the test, i.e., state the null and alternative hypotheses, the test statistic, and the rejection region.
- d) For which values of the test statistic would a decision against skier A be taken without resort to a coin toss?
- e) Perform the test. How does the test decide? Is a coin toss necessary or not?

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 $\rightarrow \alpha = 1\%$
- How many observations are necessary to reduce the width of the confidence interval by half?
- How many independent determinations of the lead content need to be planned if with 99% confidence, it is to be determined to within 1 ppb, that is, if the width of the 99% confidence interval should not be greater than 2 ppb?

We need to perform a z -test $\Rightarrow$ we want to find μ

$$a) I = \left[\bar{X} - \frac{\sigma}{\sqrt{n}} z_{1-\frac{\alpha}{2}}, \bar{X} + \frac{\sigma}{\sqrt{n}} z_{1-\frac{\alpha}{2}} \right], \text{ where } z_{1-\frac{\alpha}{2}} =$$

$$b) n = 40$$

$$c) 0.5 = \frac{\sigma}{\sqrt{n}} \cdot z_{1-\frac{\alpha}{2}} = \frac{10}{\sqrt{n}} \cdot 2.58 \Rightarrow \text{solve for } n$$

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- State the null and alternative hypotheses for the statistical test. Justify your choice.
- Provide the distribution of X under H_0 . Is the null hypothesis rejected at the 1% significance level?
- What is the probability of rejecting the null hypothesis if the true prediction probability is $p = 0.75$? $\rightarrow$ power of the test

$$a) H_0: p \leq 0.5 \quad // \quad H_1: p > 0.5$$

$$b) \text{binomcdf}(X \leq k) = 1 - \alpha$$

$$c) \Rightarrow \text{power of the test}$$

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a)

b)

c)

$$e) T = \frac{\bar{X}_n - \mu_0}{\hat{\sigma}/\sqrt{n}} \quad \text{compare with } t_{n-1, p}$$

$$X = \textcircled{k}$$



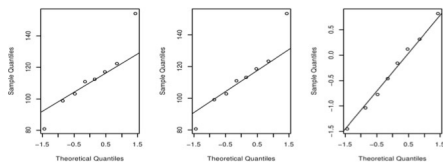
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